

Chemistry: Theory Practice Paper

Q1. (Atomic Structure) Define Heisenberg's Uncertainty Principle and explain its significance in the quantum mechanical model.

Q2. (Bonding) Explain the Hybridization process in methane (CH_4) using VSEPR theory concepts.

Q3. (Thermodynamics) Derive the relationship $\Delta G = \Delta H - T\Delta S$ and explain the condition for reaction spontaneity.

Q4. (Kinetics) Distinguish between 'Order of Reaction' and 'Molecularity'.

Why is order determined experimentally?

Q5. (Coordination) Explain Crystal Field Theory (CFT). How does it explain the color of coordination compounds?

Q6. (Organic) Describe the mechanism of $\text{S}_\text{N}2$ reaction with suitable examples and energy profile diagrams.